

9-12-19(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2019

Max. Marks: 100

Class: T.Y.

Name of the Course: DBMS

Course Code: UCEI605

Duration: 3 Hrs

Semester: VI

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q1 (a)	With a neat diagram describe the overall system structure of DBMS. OR Compare Database system and File system in Detail.	10
Q1 (b)	Give the symbols used for Specialization, Generalization, Cardinality, Relationship and Entity in ER Diagram. Explain each of them by considering 'RAILWAY SYSTEM'.	10
Q 2 (a)	List and explain the data types supported by SQL. Show the use of 'Distinct' and 'AS' keywords in SQL select statement. OR Explain DML, DDL and DCL with suitable examples.	10
Q. 2 (b)	Consider the following database schema and write nested queries (wherever applicable) in SQL. Supplier (id, name, city) Parts (pno, pname, pdescription, cost) Supply (id, pno, Quantity) a. Find the names of the parts supplied by "Ram". b. Find the names of the suppliers who supply "Nuts". c. Find the cost of the 'Bolts' being supplied by "Thane suppliers". d. Find the number of parts supplied by supplier from the city 'Mumbai'. e. Perform insertion for a new supplier with name 'Jill' from city 'Bangalore' who supplied the Part 'P3' in 100 quantities.	15

Q 3 (a)	Explain (any three): - Normalization and its need - Functional Dependency with example - Lossless Decomposition and its properties - Anomalies in Normalization - BCNF in detail	15
Q 3 (b)	Consider the following schema of student database: STUDENT (Id, Name, Dept, HOD, City, State, Pincode) Normalize the above relation up to 3NF. Show all the functional dependencies and give all assumptions very clearly.	10
Q 4 (a)	What do you mean by a Transaction? Explain desirable properties of transactions.	10
Q 4 (b)	Explain Timestamp Based Concurrency control protocols in detail. OR Explain Serializability in detail. How it can be achieved? Give the different techniques that can be used to control concurrency.	10
Q 5	What is ODBC? What is the need of ODBC? Give ODBC Architecture with the help of neat and clean diagram.	10